



CloudDrive

Technical Stack & Cloud Services

Scalable file storage on AWS · Final-year Cloud Computing project

This document lists every technology, framework and cloud service used to build CloudDrive, grouped by layer. Each entry has a custom-drawn icon and a one-line description of its role in the system.

Backend & Language

-  **Python 3.12**
Core programming language for the backend
-  **FastAPI**
High-performance async web framework (the REST API)
-  **Uvicorn**
ASGI server that runs the FastAPI app
-  **SQLAlchemy**
ORM — maps Python objects to database tables
-  **Pydantic**
Request/response validation and settings

Frontend

-  **HTML5**
Structure of the single-page dashboard UI
-  **CSS3**
Styling — responsive, modern Google-Drive-like design
-  **JavaScript (Vanilla)**
Upload flow, drag & drop, sharing, view logic

Security & Authentication

-  **JWT (python-jose)**
Stateless bearer-token authentication
-  **bcrypt**
One-way password hashing (never stored in plain text)

Storage, Database & SDK

-  **AWS S3**
Object storage for file bytes (via pre-signed URLs)
-  **PostgreSQL**
Production metadata database (users, files, shares)
-  **SQLite**
Zero-setup local database for development/demo
-  **psycopg2**
PostgreSQL driver used in production
-  **boto3**
AWS SDK for Python — generates S3 pre-signed URLs

AWS Cloud Infrastructure



Amazon S3

Versioned, encrypted, private file storage



AWS IAM

Least-privilege roles & policies for secure access



Amazon RDS

Managed PostgreSQL database (Terraform)



Amazon ECS Fargate

Serverless containers, auto-scaling 1->4 (Terraform)



Application Load Balancer

Distributes traffic across API containers



Amazon VPC

Isolated network: public/private subnets, NAT



Amazon ECR

Private Docker image registry

DevOps, IaC & Tooling



Terraform

Infrastructure as Code — builds the whole AWS stack



Docker

Containerizes the app for portable deployment



GitHub Actions

CI/CD — runs tests, builds & deploys on push



Git & GitHub

Version control and code hosting



pytest

Automated test suite (19 tests: auth, upload, sharing)

Hosting & Networking



Render

Cloud platform for permanent deployment + managed DB



Cloudflare Tunnel

Public HTTPS URL for the live demo